

Sai Akshay Damera

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Summary

Data Science Graduate student at the University of Houston with experience working on machine learning, deep learning, computer vision, and data analysis projects. Strong Python, R, and SQL skills for data preprocessing, statistical analysis, model development, and evaluation. Worked on projects involving medical image classification, predictive modeling, and computer vision, and interested in applying these skills to real-world research problems. Comfortable learning new methods, working independently, and collaborating with research teams.

Education

University of Houston
M.S. Engineering Data Science and AI
Vellore Institute of Technology
B.Tech, Engineering

Houston, TX
Aug 2025 – May 2027
Amaravathi, India
Aug 2021 – May 2025

Experience

Data Science

Jan 2025 – May 2025

Tech Autonetics

- Pull Built and maintained a weekly Power BI dashboard tracking quality and performance metrics (KPIs, defect and process data), giving the team consistent, up-to-date visibility into operational trends.
- Automated the cleaning and formatting of raw operational data using Python and SQL, replacing manual spreadsheet prep and ensuring the dashboard ran on clean, analysis-ready data every week.
- Queried and analyzed operational datasets in SQL and Excel to surface patterns and trends, translating findings into data-driven insights that supported the team's decision-making.

MERN Full Stack Developer

Aug 2023 – Nov 2023

Ethnus

Hyderabad, India

- Built responsive UI components and pages for a gaming community platform using React.js, including user authentication/profile pages, a game browsing and listing interface, and a community forum for member discussions.
- Integrated the frontend with backend REST APIs (Node.js, Express.js) to fetch and render live data — user profiles, game listings, and forum posts — styled with CSS/SCSS for a consistent, responsive UI.
- Collaborated within an Agile team, participating in sprint planning and code reviews to ship features iteratively across the platform.

Business Analyst Intern

Jun 2024 – Aug 2024

Finlatics

Hyderabad, India

- Conducted market research and competitor analysis across business projects, benchmarking industry trends and competitor strategy to inform business positioning.
- Performed financial risk assessments to evaluate potential exposure on ongoing projects and identify mitigation strategies.
- Prepared reports and recommendations that translated findings into clear, actionable guidance for leadership decision-making.

Projects

YOLOv10 Leukemia Detection - Deep Learning Medical Image Classification | Python, YOLOv10, Deep Learning, Computer Vision

- Fine-tuned a YOLOv10 model to detect and classify leukemia subtypes from blood-smear images, applying image preprocessing and evaluating performance using precision, recall, and mean Average Precision (mAP).
- Built an end-to-end image classification workflow covering dataset preparation, model training, validation, and performance analysis to assess the model's effectiveness across leukemia classes.

Medical Insurance Cost Prediction | Predictive Analytics, Machine Learning

- Analyzed 1,338 medical insurance records and compared Linear Regression, Forward Stepwise Regression, and Random Forest models using MSE, RMSE, and R²; the tuned Random Forest achieved the best RMSE of ~\$4,398.
- Performed EDA, data-quality checks, categorical encoding, correlation analysis, and visualization to identify key drivers of insurance charges, with smoking status emerging as the strongest predictor.

Image-Based Attendance System | Real-Time Computer Vision Application

- Built a real-time facial-recognition attendance system using Python and OpenCV, integrating database functionality and a user interface to automate attendance tracking, reporting, and record retrieval.
- Implemented an image-processing and face-matching workflow to identify registered users and automatically record attendance, reducing reliance on manual classroom attendance processes.

Few-Shot Voice Classification - Speaker Identification with Limited Data | Python, Deep Learning, Few-Shot Learning, Audio Processing

- Developed a few-shot learning model to classify speakers using a limited number of audio samples, applying audio preprocessing and neural-network-based feature extraction.
- Designed a voice-matching pipeline to compare learned speaker representations and support speaker identification under limited-data conditions.

Technical Skills

Programming & Data: Python, R, SQL, Java, JavaScript; Pandas, NumPy, data preprocessing, exploratory data analysis (EDA), statistical analysis.

Machine Learning & AI: Deep Learning, Computer Vision, YOLOv10, Few-Shot Learning, predictive modeling, regression, Random Forest.

Computer Vision & Analysis: OpenCV, facial recognition, image preprocessing, feature extraction, Power BI, Tableau, Advanced Excel.

Databases & Development: MySQL, MongoDB, HTML, CSS, React.js, Node.js, Express.js; database management and data validation.

Strengths: Research-oriented problem solving, analytical thinking, data interpretation, technical documentation, collaboration, adaptability.